Chapter 5

5. Classification.

5.1. Bayes discriminant rule.

5.2. Classification of two normal populations.

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Introduction

Classification is a very common and important problem in a data analysis.

The objective is to assign each observation X (a p-vector) in the sample to a certain population.

It may be also viewed as a regression problem, where the response variable Y is qualitative (categorical data).

Introduction

Examples:

- A patient arrives at the emergency room with a certain health condition X (e.g. blood pressure, hart rate, etc.) and we want to assign a medical diagnoses Y (e.g. stroke yes/no).
- More sophisticated version: we have an image X of a birthmark and want to figure out if it is dangerous or not (Y again a zero-one variable).
- An autonomous car gets input X from a number of sensors. The task is to classify which action to take. E.g.

$$Y \in \{\text{stop, turn left, turn right, go straight}\}.$$

Introduction

Classification is also referred to as 'supervised learning".

In classification problems the different populations are assumed to be known (in contrast to cluster analysis—unsupervised learning).

We will discuss different approaches for classification. We first introduce statistical methods

1. Bayes discriminant rule;
2. linear discriminant analysis;
3. logistic regression.

At the end of the section we also introduce a few algorithmic methods. kNN classification, decision trees, bagging, random forests.

Introduction

Let us state the problem in mathematical terms. To begin with, we consider two populations π_1 and π_2, which are associated with two density functions f_1 and f_2 (over $\mathbb{R}^p$).

The two populations are different by location, dispersion or some other characteristic.

Introduction

We consider the following problem:

Given the realization $x = (x_1, \ldots, x_p)'$ of a p-random vector $X = (X_1, \ldots, X_p)'$ coming either from population π_1 or π_2 (i.e., $X \sim f_1$ or $X \sim f_2$), how shall we attribute x to one of the two populations?

$\rightsquigarrow$ A classification rule consists of partitioning χ (the possible values of X) into $\chi = R_1 \cup R_2$ where the disjoint regions (R_1, R_2) are such that

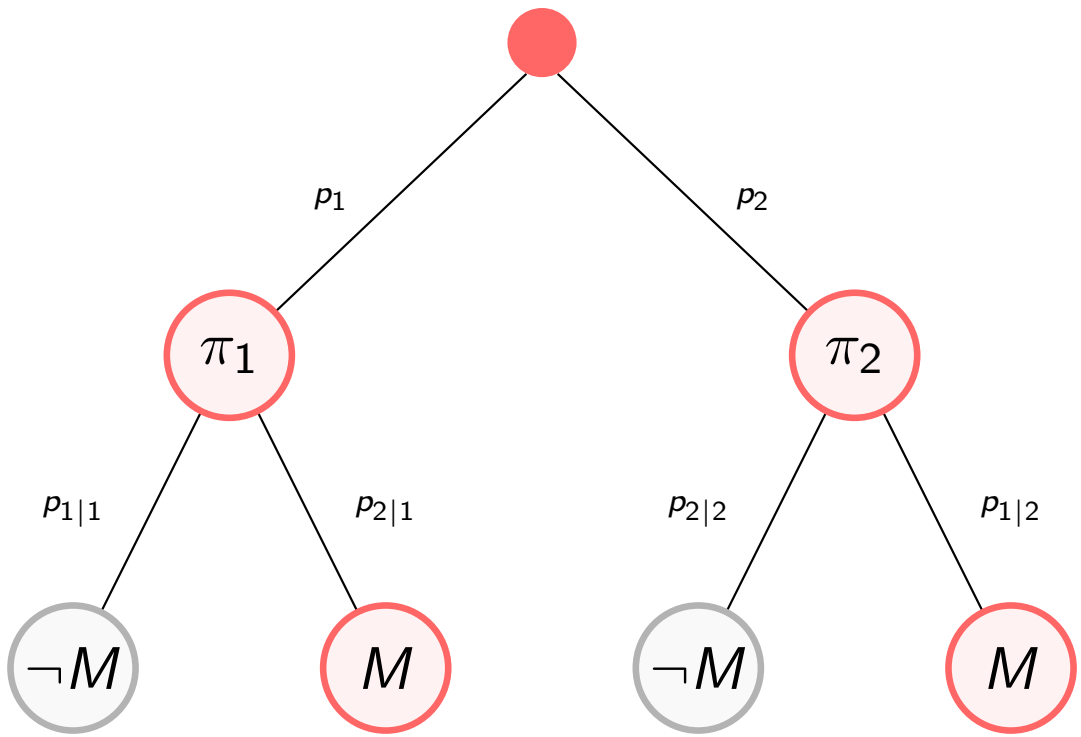
- x is classified in π_1 if $x \in R_1$ and
- x is classified in π_2 if $x \in R_2$.

Bayes discriminant rule

Let us assume prior probabilities p_1, p_2. Hence X was sampled from π_i with probability p_i.

Let p_{mc} be the probability of misclassifying X. It is given by

$$p_{\text{mc}} = p_{2|1} \times p_1 + p_{1|2} \times p_2.$$



Bayes discriminant rule

Here we have

$$p_{i|j} = P[X \in R_i | X \sim f_j] = \int_{R_i} f_j(x)\, dx.$$

A sensible classification procedure is to choose R_1 (and then $R_2 = \chi \backslash R_1$) such that p_{mc} is minimized.

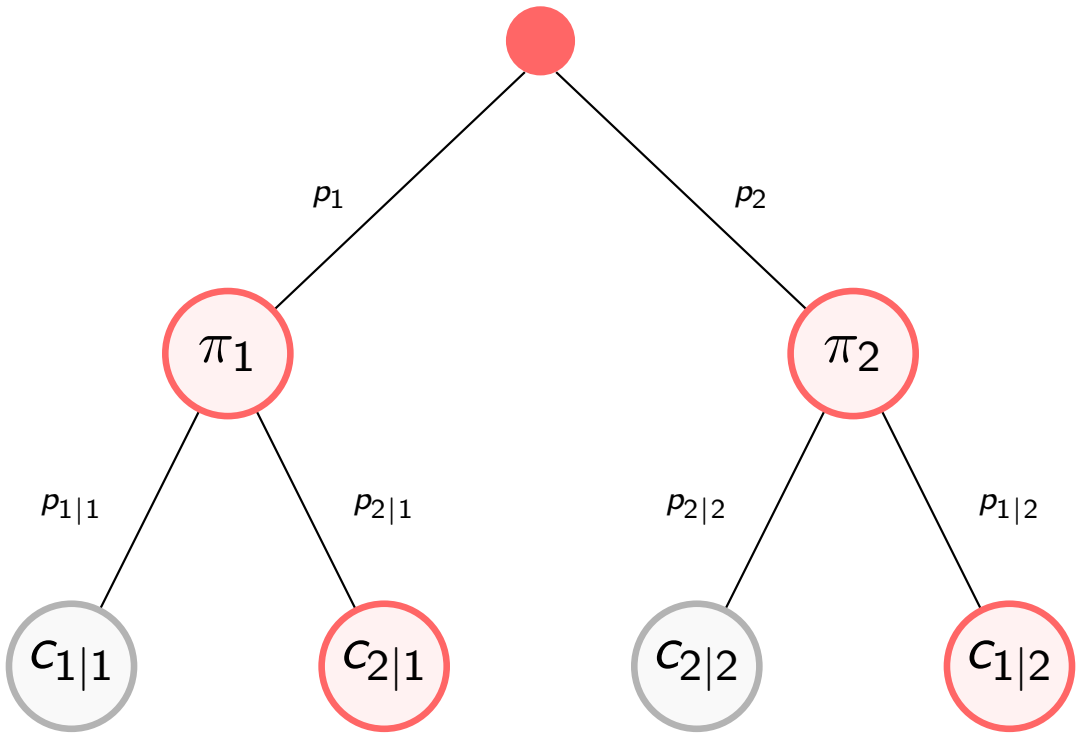
Bayes discriminant rule

This idea can be generalized by taking into account that there are potentially different costs of misclassification.

Let $c_{1|2}$ and $c_{2|1}$ be the costs for classifying an observation π_1 though actually comes from π_2 and vice versa, then a classification procedure should determine R_1 such that the average misclassification costs are minimized:

$$\begin{aligned} E_{\text{mc}} &= E[\text{costs}|X \sim f_1]\, p_1 + E[\text{costs}|X \sim f_2]\, p_2 \\ &= \left(0 \times p_{1|1} + c_{2|1}\, p_{2|1}\right) p_1 + \left(c_{1|2}\, p_{1|2} + 0 \times p_{2|2}\right) p_2 \\ &= c_{2|1}\, p_{2|1}\, p_1 + c_{1|2}\, p_{1|2}\, p_2. \end{aligned}$$

Bayes discriminant rule



Bayes discriminant rule

The following result provides the optimal classification procedure (R_1, R_2) in this general context:

Theorem

The classification procedure (R_1, R_2) which minimizes the average misclassification costs is given by

$$R_1 = \left\{ x \in \chi \,\middle|\, \frac{f_1(x)}{f_2(x)} \geq \frac{c_{1|2}\, p_2}{c_{2|1}\, p_1} \right\} \quad and \quad R_2 = \chi \backslash R_1.$$

Bayes discriminant rule

Proof.

$$\begin{aligned} E_{\text{mc}} &= c_{2|1}\, p_{2|1}\, p_1 + c_{1|2}\, p_{1|2}\, p_2 \\ &= c_{2|1}\,(1 - p_{1|1})\, p_1 + c_{1|2}\, p_{1|2}\, p_2 \\ &= c_{2|1}\, p_1 + (c_{1|2}\, p_{1|2}\, p_2 - c_{2|1}\, p_{1|1}\, p_1) \\ &= c_{2|1}\, p_1 + \int_{R_1} (c_{1|2}\, f_2(x)\, p_2 - c_{2|1}\, f_1(x)\, p_1)\, dx \end{aligned}$$

E_{mc} is minimal if the integrand is negativ for all $x \in R_1$. □

Bayes discriminant rule

Remarks:

- It suffices to know the ratios $\frac{f_1}{f_2}$, $\frac{c_{1|2}}{c_{2|1}}$ and $\frac{p_2}{p_1}$ to determine the optimal classification.
- If the probabilities p_1, p_2 are equal or unknown,

$$R_1 = \left\{ x \in \chi \,\middle|\, \frac{f_1(x)}{f_2(x)} \geq \frac{c_{1|2}}{c_{2|1}} \right\} \quad \text{and} \quad R_2 = \chi \backslash R_1.$$

- If the costs $c_{1|2}$, $c_{2|1}$ are equal or not specified,

$$R_1 = \left\{ x \in \chi \,\middle|\, \frac{f_1(x)}{f_2(x)} \geq \frac{p_2}{p_1} \right\} \quad \text{and} \quad R_2 = \chi \backslash R_1.$$

- If p_1, p_2, $c_{1|2}$, $c_{2|1}$ are not specified,

$$R_1 = \left\{ x \in \chi \,\middle|\, \frac{f_1(x)}{f_2(x)} \geq 1 \right\} \quad \text{and} \quad R_2 = \chi \backslash R_1.$$

(Maximum likelihood discriminant rule.)

Classification of two normal populations

Let us consider $\pi_i = \mathcal{N}_p(\mu_i, \Sigma_i)$, $i = 1, 2$, with $\Sigma_1 = \Sigma_2 (=: \Sigma)$.

Proposition

Let $a := \Sigma^{-1}(\mu_1 - \mu_2)$. Then the optimal classification procedure classifies x in π_1 iff

$$d^2_\Sigma(x, \mu_2) - d^2_\Sigma(x, \mu_1) \geq 2 \log \left[\frac{c_{1|2}\, p_2}{c_{2|1}\, p_1} \right],$$

or iff

$$a'x \geq a' \frac{\mu_1 + \mu_2}{2} + \log \left[\frac{c_{1|2}\, p_2}{c_{2|1}\, p_1} \right]$$

and in π_2 otherwise.

The resulting classification is a linear discriminant rule.

Classification of two normal populations

Proof: It follows from the general theorem that classifying x in π_1 is optimal if $\log\left[\frac{f_1(x)}{f_2(x)}\right] \geq \log\left[\frac{c_{1|2}\, p_2}{c_{2|1}\, p_1}\right]$. Hence

$$\log\left[\frac{f_1(x)}{f_2(x)}\right] = -\frac{1}{2}\underbrace{(x-\mu_1)'\Sigma^{-1}(x-\mu_1)}_{d^2_\Sigma(x,\mu_1)} + \frac{1}{2}\underbrace{(x-\mu_2)'\Sigma^{-1}(x-\mu_2)}_{d^2_\Sigma(x,\mu_2)}$$

$$= (\mu_1-\mu_2)'\Sigma^{-1}x - \frac{1}{2}(\mu_1'\Sigma^{-1}\mu_1 - \mu_2'\Sigma^{-1}\mu_2) = a'x - a'\frac{\mu_1+\mu_2}{2}.$$

□

Classification of two normal populations

Remark: Note that if $\log\left[\frac{c_{1|2}\, p_2}{c_{2|1}\, p_1}\right] = 0$, then the classification rule is based on

$$a'\left(x - \frac{\mu_1 + \mu_2}{2}\right) \geq 0.$$

The decision boundary is the hyperplane

$$a'\left(x - \frac{\mu_1 + \mu_2}{2}\right) = 0.$$

Classification of two normal populations

In practice, μ_1, μ_2 and Σ are unknown and we have to base our decision on a training sample of independent observations $X_1, \ldots, X_{n_1} \sim \mathcal{N}_p(\mu_1, \Sigma)$ and $Y_1, \ldots, Y_{n_2} \sim \mathcal{N}_p(\mu_2, \Sigma)$.

The empirical classification rule classifies x in π_1 iff

$$\underbrace{(\bar{X} - \bar{Y})' S_{\text{pool}}^{-1}}_{\hat{a}'} x \geq \underbrace{\frac{1}{2}(\bar{X} - \bar{Y})' S_{\text{pool}}^{-1}(\bar{X} + \bar{Y}) + \ln\left[\frac{c_{1|2}\, p_2}{c_{2|1}\, p_1}\right]}_{\widehat{T}}.$$

Classification of two normal populations

Let us now consider the case where $\pi_i = \mathcal{N}_p(\mu_i, \Sigma_i)$, $i = 1, 2$, with unequal variances Σ_i. In this case the quadratic terms don't cancel. We obtain again a rule of the form

$$\text{classify } x \text{ in } \pi_1 \text{ iff} \quad d^2_{\Sigma_2}(x, \mu_2) \geq d^2_{\Sigma_1}(x, \mu_1) + \text{threshold}.$$

Working out this expression we obtain the following quadratic classification rule (exercise):

Classification of two normal populations

Proposition

Let

$$t := \frac{1}{2}\left(\log\left[\frac{|\Sigma_1|}{|\Sigma_2|}\right] + (\mu_1'\Sigma_1^{-1}\mu_1 - \mu_2'\Sigma_2^{-1}\mu_2)\right) + \log\left[\frac{c_{1|2}\, p_2}{c_{2|1}\, p_1}\right].$$

Then the optimal classification procedure classifies x in π_1 if

$$-\frac{1}{2}x'(\Sigma_1^{-1} - \Sigma_2^{-1})x + (\mu_1'\Sigma_1^{-1} - \mu_2'\Sigma_2^{-1})x \geq t.$$

and in π_2 else.

Fisher's discriminant rule

Let us go back to the beginning of the problem with $\Sigma_1 = \Sigma_2$ and try to motivate the choice of the vector a more generally. We suppose $\frac{c_{1|2}\, p_2}{c_{2|1}\, p_1} = 1$.

We transform the observation X to

$$t_a(X) = \frac{a'X}{\sqrt{a'\Sigma a}}$$

(hence to a scalar) for some generic $a \in \mathbb{R}^p$.

Note that $\sqrt{a'\Sigma a} = \mathrm{sd}(a'X)$ for $X \sim f_1$ and $X \sim f_2$.

Classification of two normal populations

A potentially sensible rule is to classify X in π_1 if

$$|t_a(\mu_1) - t_a(X)| \leq |t_a(\mu_2) - t_a(X)|.$$

The question is what is a reasonable choice of a?

A natural choice is a such that $|t_a(\mu_1) - t_a(\mu_2)|$ is maximal. One can show that this is the case with

$$a = \Sigma^{-1}(\mu_1 - \mu_2),$$

a term we have seen earlier!

Some further steps show, that we end up with the same rule as in the case with two normal distributions with equal variance.

Fisher's discriminant rule

Remark 1: The function $x \mapsto a'x - \frac{1}{2}a'(\mu_1 + \mu_2)$ on which the classification rule is based, is called Fisher's linear discriminant function. We are talking about linear discriminant analysis.

Remark 2: This rule is sensible without assumption of normality. For the normal case, it coincides with the Bayes rule.

Real data illustration

The objective of this example is to classify Canadian and Alaskan salmon.

The fish have an interesting living cycle. They are born in rivers with freshwater and swim to the ocean after some years before returning to the rivers.

When they are caught in the ocean it is not clear at which place they are born, i.e. whether they belong to Canada or Alaska.

A way to discriminate is given by the growthrings (diameter of first-year freshwater growth and first-year marine growth), which tend to be bigger for Canadian fish.

Real data illustration

We have 100 observations. 50 fish from Alaska and 50 from Canada.

```
# The data has four variables:
# V1: 1=Alaska 2=Canada
# V2: 1=female 2=male
# V3: diameter-freshwater
# V4: diameter-marine


> head(salmon)
  V1 V2  V3  V4
1  1  2 108 368
2  1  1 131 355
3  1  1 105 469
4  1  2  86 506
5  1  1  99 402
6  1  2  87 423
```

Real data illustration

```
# Convert V1 to a factor and give the levels new names
> salmon$V1 <- as.factor(salmon$V1)
> levels(salmon$V1) <- c("Alaska", "Canada")

# Count how many in each group
> table(salmon$V1)

Alaska Canada
    50     50
```

Real data illustration

Plot the data:

```
> plot(salmon$V3, salmon$V4,
     bty="n",
     xlab="freshwater",
     ylab= "marine",
     # here R is ok with labels "Alaska" and "Canada" instead of 1 and 2
     col=salmon$V1
     )
> legend(
    x ="topright",
    legend = paste("Origin:", levels(salmon$V1)),
    # here we need to convert back to 1 and 2
    col = (levels(salmon$V1) <- c(1,2)),
    pch=19
  )
```

Real data illustration

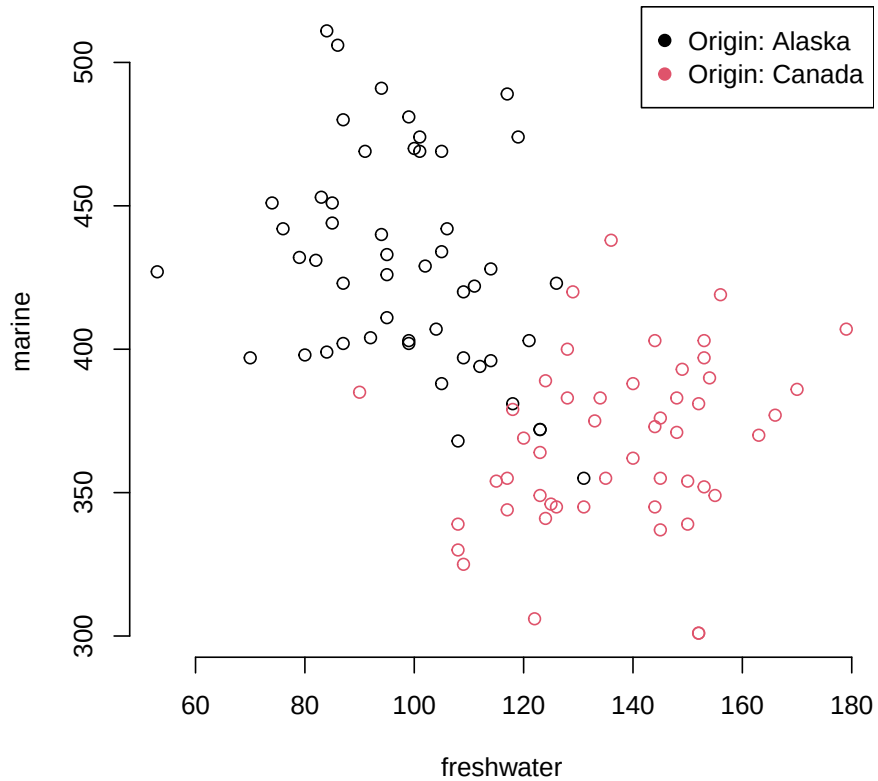


Figure 1: Alaskan fish and Canadian fish growthring diameters for a sample of 50 each.

Real data illustration

A bivariate normal seems plausible.

We do classification assuming homogenous variances $\Longrightarrow$ Fisher's LDA.

We take a training set of size 60 and evaluate on a test set of size 40.

```
> train <- sample(1:100,60)
> table(salmon$V1[train])

Alaska Canada
    30     30
```

The number of fish from Alaska and Canada happen to be balanced in sample.

Real data illustration

We perform LDA and predict the classes of the remaining fish.

```
> salmon.lda
Call:
lda(salmon$V1 ~ salmon$V3 + salmon$V4, prior = c(1, 1)/2, subset = train)

Prior probabilities of groups:
Alaska Canada
   0.5    0.5

Prior probabilities of groups:
Alaska Canada
   0.5    0.5

Group means:
       salmon$V3 salmon$V4
Alaska  99.26667  426.9333
Canada 136.80000  362.1333

Coefficients of linear discriminants:
                   LD1
salmon$V3  0.03965243
salmon$V4 -0.01835599
```

Real data illustration

The `lda` function in `R` uses

$$\text{LD1} = \frac{\hat{a}}{\sqrt{\hat{a}' S_{\text{pool}} \hat{a}}}$$

with $\hat{a} = S_{\text{pool}}^{-1}(\hat{\mu}_1 - \hat{\mu}_2)$.

Classification: classify π_1 if

$$\text{LD1}' x \geq \text{LD1}' \frac{\hat{\mu}_1 + \hat{\mu}_2}{2} + \frac{1}{\sqrt{\hat{a}' S_{\text{pool}} \hat{a}}} \log \left[\frac{p_2}{p_1} \right]$$

Real data illustration

```
> classifications <- predict(salmon.lda, salmon)$class[-train]
> true.origin <- salmon$V1[-train]
> table(classifications,true.origin)
               true.origin
classifications Alaska Canada
         Alaska     18      1
         Canada      2     19
```

We have 2 misclassification.

- $2\times$ we classified a fish from Alaska as Canadian.
- we never classified a fish from Canada as Alaskan.

Real data illustration

In the previous analysis we have used only a limited amount of data for estimation. This was done to have an out-of-sample estimate of the misclassification probabilities.

Taking the same data set for estimation and evaluation (in-sample estimate) leads to underestimate the misclassification probability.

In order to better use the amount of data and to still evaluate the quality of the classification procedure, we will use the leave-one-out procedure.

- Take all but one observation in order to estimate the procedure (e.g. S_x, S_y, ...)
- Classify the left-out observation.
- Repeat the same procedure for all observations.

From this we can estimate the misclassification probability.

Real data illustration

```
> salmon.lda.cv <- lda(salmon$V1 ~salmon$V3+salmon$V4,
      prior = c(1,1)/2, CV=TRUE)
> classifications <- salmon.lda.cv$class
> true.origin <- salmon$V1
> table(classifications,true.origin)

               true.origin
classifications Alaska Canada
         Alaska     44      1
         Canada      6     49
```

Remark: It turns out that Canadian salmon are less often misclassified. So in a certain sense the procedure is 'biased. In practice such an outcome may not be appreciated ...

Classification for $m \geq 3$ populations

Now we consider the case of m populations π_i $(i = 1, \ldots, m)$, associated to some densities f_i in $\mathbb{R}^p$ $(i = 1, \ldots, m)$.

We look for a classification rule which can attribute an observation x of a p-random vector X to its population π_i, $1 \leq i \leq m$.

Similar as for populations, the classification rule is based on partitioning χ into sets

$$(R_i,\ i = 1, \ldots, m)$$

and to classify x in π_i iff $x \in R_i$.

Classification for $m \geq 3$ populations

Let us denote

- $c_{i|j}$ the cost of classifying an object of π_j in π_i,
- p_j the a priori probability that X comes from π_j
- $p_{i|j} := P[X \in R_i | X \sim f_j] = \int_{R_i} f_j(x)\, dx$ the conditional that an object is classified in π_i given that it comes from π_j.

A classification procedure $(R_i,\ i = 1, \ldots, m)$ is called optimal, if it minimizes the average cost of misclassifcation, which can be written as (with $c_{i|i} = 0$):

$$E_{\text{mc}} = \sum_{j=1}^{m} E[\text{cost} | X \sim f_j]\, p_j = \sum_{j=1}^{m} \left[\sum_{i=1}^{m} c_{i|j}\, p_{i|j} \right] p_j.$$

Classification for $m \geq 3$ populations

One can show the following result:

Theorem

The procedure which minimizes the mean misclassification costs, consists in classifying x in population π_i for which

$$h_i(x) := \sum_{j=1}^{m} c_{i|j}\, p_j\, f_j(x)$$

is minimal.

Classification for $m \geq 3$ populations

Remarks:

- This result extends the case $m = 2$;
- If the $c_{i|j}$ are equal or not specified, the optimal classification rule attributes x to the population π_i for which $p_i f_i(x) = \max_j \{p_j f_j(x)\}$.

The last result we can interpret with Bayes' theorem. If we think of $f_i(x) = P(X = x|\pi_i)$ (if $f_i(x)$ were a probability mass function) then by Bayes' theorem

$$P(\pi_i|X = x) = \frac{P(X = x|\pi_i)P(\pi_i)}{P(X = x)} = \frac{f_i(x)p_i}{P(X = x)}.$$

Classification for $m \geq 3$ populations

Proof of the Theorem:

$$
\begin{aligned}
E_{\text{mc}} &= \sum_{j=1}^{m} \left[\sum_{i=1}^{m} c_{i|j}\, p_{i|j} \right] p_j \\
&= \sum_{i=1}^{m} \sum_{j=1}^{m} c_{i|j}\, p_j \int_{R_i} f_j(x)dx \\
&= \sum_{i=1}^{m} \int_{R_i} h_i(x)dx.
\end{aligned}
$$

□

Classification for $m \geq 3$ populations

Let us consider the case where $\pi_i = \mathcal{N}_p(\mu_i, \Sigma_i)$, $i = 1, \ldots, m$. (We don't include misclassification costs).

Proposition

Let

$$d_j(x) := -\frac{1}{2}\log|\Sigma_j| - \frac{1}{2}(x-\mu_j)'\Sigma_j^{-1}(x-\mu_j) + \log p_j.$$

Then the optimal classification procedure classifies x in π_i iff $d_i(x) = \max_j\{d_j(x)\}$.

Proof. Exercise.

In practice all the involved means and covariances are replaced by the respective estimates.

Logistic regression

Another common classification rule when we have to distinguish between 2 populations is by logistic regression.

We consider hence again a random vector X coming either from population π_1 or population π_2.

We can think of an observation $(Y, X')'$ with $Y = 0, 1$ depending on whether X is from population π_1 or population π_2, respectively.

We would like to model the conditional probabilities $P(Y = 1|X)$.

In the logistic regression we assume a model of the form

$$E(Y|X) = P(Y = 1|X) = g(X'\beta)$$

with the logistic response function $g(t) = \frac{e^t}{1+e^t} \in (0, 1)$.

Logistic regression

Conditional on the sample $x_1, \ldots, x_n$ the likelihood function for $Y_1, \ldots, Y_n$ is given by

$$\prod_{i=1}^{n} g(x_i'\beta)^{Y_i}(1 - g(x_i'\beta))^{1-Y_i}.$$

We have

$$\hat{\beta} = \text{argmax}_\beta \prod_{i=1}^{n} g(x_i'\beta)^{Y_i}(1 - g(x_i'\beta))^{1-Y_i}.$$

We use R in order to estimate the model (maximize likelihoodfunction w.r.t. β).

For given β we can calculate $P(Y = 1|X = x)$. If this probability is $> .5$ we will classify X in population π_2.

Logistic regression

```
> salmon.glm <- glm(V1~V3+V4, family=binomial, data=salmon, subset = train)
> summary(salmon.glm)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) 11.14106    9.17589   1.214  0.22468
V3           0.10383    0.03964   2.620  0.00880 **
V4          -0.06055    0.02218  -2.731  0.00632 **
---
Signif. codes:  0 *** 0.001 ** 0.01 * 0.05 . 0.1  1
```

The estimate is $\hat{\beta} = (11.141, 0.104, -0.061)$. So if x_1 is the freshwater and x_2 the marine diameter of a fish, we calculate

$$\hat{P}(\text{fish belongs to } \pi_2) = g(11.141 + 0.104x_1 - 0.061x_2).$$

Logistic regression

We can now make out-of-sample predictions for the probability of Y being 1, given the value of X.

```
> # out of sample prediction (beta x)
> salmon.glm.aos <- predict(salmon.glm,salmon)[-train]
> # estimated probabilities (g(beta x))
> probs <- exp(salmon.glm.aos)/(1+exp(salmon.glm.aos))
> preds <- character(length = 40)
> # attributing class
> for(i in 1:40){if(probs[i]<1/2) preds[i]<-"Alaska" else preds[i]<-"Canada"}
> true.origin <- salmon$V1[-train]
> table(preds,true.origin)
        true.origin
preds    Alaska Canada
  Alaska     17      1
  Canada      2     20
```

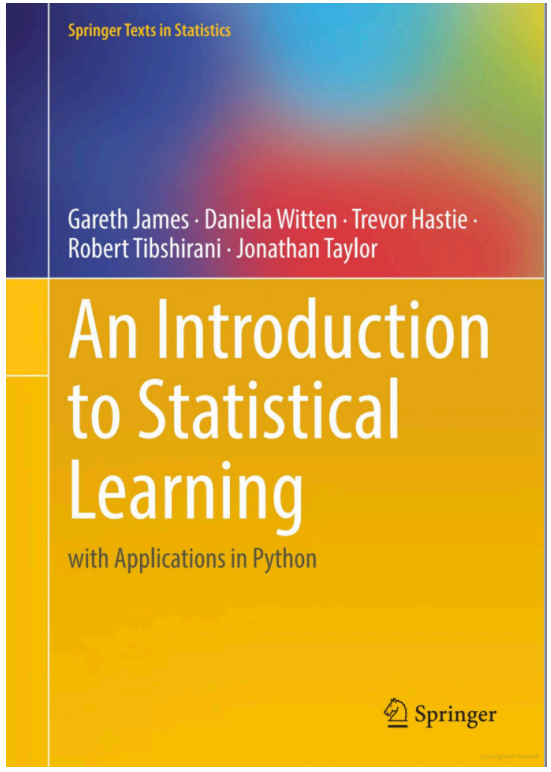
Model-free techniques

Up to this point we have been using statistical models for the explanatory variables in order to motivate our classification procedures. The purpose of this section is to give a short introduction in some model-free techniques, which are often used in the machine learning communities. These techniques have in common that they are:

1. purely computational;
2. no model assumptions are required;
3. can *sometimes* largely outperform statistical models in terms of predictive power.

Model-free techniques

An excellent survey is given in the book:



A price to pay is often lack of interpretability!

Model-free techniques

One of the simplest approaches is the K-nearest neigbours (KNN) approach. The idea is rather simple:

We consider a training set $\{(y_i, x_i) : 1 \leq i \leq n\}$ where $y_i \in \{c_1, \ldots, c_m\}$ is the categorical variable of interest and $x_i \in \mathbb{R}^p$ are our explanatory variables. (Since we don't run a stochastic approach, we use lower case letters.)

To classify a newly observed x_0, proceed as follows:

1. compute distances $d_i = d(x_0, x_i)$ for $1 \leq i \leq n$;
2. let $\mathcal{N}_0$ be the indices related to the K smallest values d_i;
3. the prediction is the class c_ℓ which is most often contained in the vector $(y_j : j \in \mathcal{N}_0)$.

E.g. when $K = 1$, the predicted class is the class of the nearest neighbour of x_0 in the training data.

Model-free techniques

Remarks:

- We need to provide K; can e.g. be obtained via a cross-validation.
- Different K can lead to very different results. Larger K leads to smoother *decision boundary*.
- Is able to cope with highly non-linear *decision boundary*. In this case it has good potential to outperform the model-based methods, which are boiling down to linear decision boundaries.
- Is good for large training samples. Particularly, when p is large, the training set must be very large! (Curse of dimensionality.) Think of the p-dimensional unit cube with a fixed number of n points. When p is getting larger, we will have no close elements.

Model-free techniques

```
> library("class")
> training <- salmon[train,]
> test <- salmon[-train,]
> knn.preds <- knn(training[,3:4], test[,3:4],cl=salmon$V1[train],k=1)
> table(knn.preds,true.origin)
          true.origin
knn.preds Alaska Canada
   Alaska     16      2
   Canada      3     19
```

Model-free techniques

Another common technique is based on decision trees. The idea is again to partition the predictor space into J regions $R_1, \ldots, R_J$ and to make the same classification for all $x \in R_k$.

Note that J can be larger than m. Then we will have several regions whose elements are attributed to the same class.

The question is how the segmentation into the J regions should be done. A common approach is to use a number of splitting rules, which can be summarised in a tree.

Model-free techniques

- In theory the regions can have any shape, but in practice rectangles/boxes are the basic building blocks.
- Suppose we have already $j-1$ regions. We will split one of them to get j regions. For all elements in one region we give the same predicted class. W.l.o.g. we assume that we split Region R_{j-1}. We then get

$$R_1, \ldots, R_{j-2}, R_{j-1,1}, R_{j-1,2}.$$

  We define this as our new regions $R_1, \ldots R_j$.
- The split is of the form $R_{j-1,1} = \{x \in R_{j-1} : x_\ell \leq s\}$ and $R_{j-1,2} = \{x \in R_{j-1} : x_\ell > s\}$.

Model-free techniques

- To see if the split is good, let us define $\hat{p}_{\ell|k}$ to be the proportion of training observations in R_ℓ coming from class k.
- The objective is to have all $\hat{p}_{\ell|k}$ close to 0 or 1 for the m's correpsonding to our two new regions obtained by the splitting. If $\hat{p}_{\ell|k} \approx 1$ then most training data in R_ℓ belong to class c_k. This motivates to classify a new test observation in R_ℓ also as class c_k.

Model-free techniques

- Common measures which aggregate $\hat{p}_{\ell|1}, \ldots, \hat{p}_{\ell|m}$ are the Gini index G or the cross-entropy D:

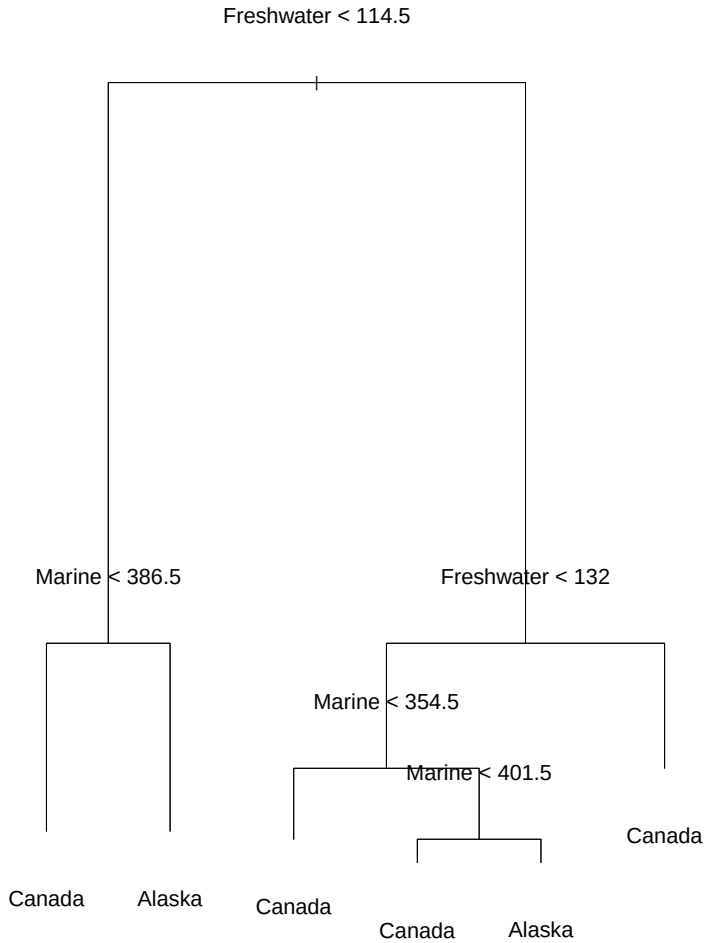
$$G = \sum_{k=1}^{m} \hat{p}_{\ell|k}(1 - \hat{p}_{\ell|k}) \quad \text{and} \quad D = -\sum_{k=1}^{m} \hat{p}_{\ell|k} \log \hat{p}_{\ell|k}.$$

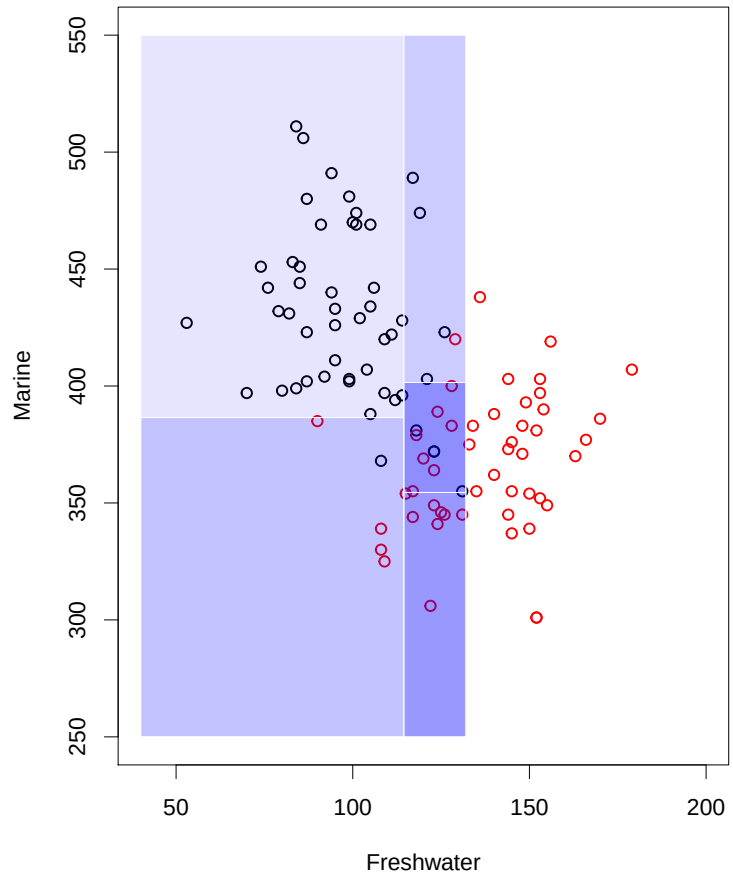
  These sums are small if all $\hat{p}_{\ell|k}$ are close to zero or one.
- After some splitting, add up all G's or D's from each region. The split leading to smallest number wins.
- We continue the splitting in the same manner with the new regions until a stopping criterion is reached. (E.g. stop if no region contains more than 10 elements.)

Model-free techniques

```
> library("tree")
> names(salmon) <- c("origin", "m/f", "freshwater", "marine" )
> salmon.tree=tree(origin~freshwater+marine,data=salmon,method="class")
> plot(salmon.tree)
> par(cex=1)
> text(salmon.tree,pretty = 0)
```

Model-free techniques





Model-free techniques

A decision tree (like any classification technique) should be such that it gives good out-of-sample performance. If we grow the tree on the original data as we did above, it may fit very well, but give bad out-of-sample predictions.

This is because often trees become too complex (overfitting). A way out would be to grow a smaller tree. (This has additional advantage of easier to interpretation.) A way to do this, is to stop growing the tree if the decrease in G exceeds a threshold (i.e. as soon as we fail to have some considerable improvement).

The problem with this approach is that a good split may only occur deep in the tree; something that we miss if we just stopped because the current improvement was poor.

Model-free techniques

Tree pruning is a method where we first grow a large tree, and then cut back to a subtree. Different algorithms exist, which we don't investigate here further.

One approach is to do cross-validation. E.g. we try on a test set which subtree gives the best prediction. However, this is numerically very intense and only feasible if the tree is comparably small. This is because the number of subtrees (depending on the structure of the tree) may grow at super-exponetial rate. In a complete tree the number of subtrees A_n at depth n satisfies the recursion (exercise)

$$A_n = A_{n-1}^2 + 1.$$

Model-free techniques

Some advantages and disadvantages of trees:

- Trees are very easy to explain (even easier than regression).
- Easy graphical representation.
- Easy to include qualitative predictors.
- Often not so good prediction quality.
- Very non-robust: changing the data slightly, may lead to entirely different tree.

We now show how to overcome the two problems mentioned above. The price to pay will be at the cost of interpretability.

Bagging

Bagging is a technique which is based on bootstrapping and can be used in different context. Roughly, the purpose is to reduce variability of predictions.

A general statistical phenomenon is the bias-variance trade-off. If an estimator or predictor has little bias, it often comes with large variance. On the other hand, if we try to reduce variability we increase the bias.

This is also true for our decision trees. They tend to have high variance. If we randomly split the data into two halves and fit to each half a separate tree, we might get quite different results.

Bagging

1. We bootstrap n (i.e. taking a random sample from the data, with replacement—see Chapter 7) times with replacement from our sample. This yields:

$$x_{11}^*, \ldots, x_{1n}^*.$$

2. Repeating this B times, we get B bootstrap samples.

$$(x_{i1}^*, \ldots, x_{in}^*), \quad 1 \leq i \leq B.$$

3. On each bootstrap training set we derive a classification procedure as explained previously. When we construct decision trees we will not prune in order to get low bias. We accept high variance.
4. So for each x we have B prediction algorithms. To reduce again the variance, we do a majority-vote: choose the class which has been predicted most often.

Bagging

Note that there is a relatively easy way to get valid out-of-sample classification error rates. We choose out-of-bag (OOB) data (those that were not in the bootstrap sample) for predicting the class and from this the error.

More precisely: take some x_i. Consider all bootstrap samples which didn't contain x_i. With these samples we build classification trees and perform the bagging (i.e. majority vote for predicted class) and predict the class of x_i. Since x_i has not been used for estimation, we get an out of sample prediction. Repeat this for all i and calculate classification errors.

Requires only calculating B trees, while e.g. with leave-one-out CV we need to grow nB trees.

Random forests

Random forests work similar as bagging. We create again bootstrap samples, and majority votes on the decisions coming from the B decision trees from each bootstrap sample.

The difference is that for each tree we are not considering all p components for the splitting. Rather we choose for each bootstrap sample only m (typically $m \approx \sqrt{p}$) components at random, on which we do the splitting.

The rational behind this approach is as follows: if we proceed as in bagging, then the predictions in the trees tend to be correlated: an important split will occur in many bootstrap samples. In the extreme case we get always the same predictions with all trees. Then averaging doesn't lower the variance. With the new approach, the hope is to get a lower correlation between predictions.